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Concept Note

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Nuclear energy routing

Making nuclear energy pathways microscopically controllable

The opportunity: from electronics to nucleonics

Electronics became transformative when charge and electronic excitations could be switched, routed and converted with microscopic precision. Nuclear systems offer no comparable toolbox today. Their energy is largely released through pathways determined by nuclear structure and reaction dynamics, with little opportunity to intervene. What if we could influence where nuclear energy goes and how it is released?

Such control could open new possibilities for nuclear technology, from controlled excitation and release in nuclear batteries to medical isotope production and nuclear waste remediation. The most ambitious application is nuclear reaction engineering: creating alternative pathways through which nuclear reactions can release their energy, seeking to avoid the most damaging reaction products. In fusion, such control could reduce neutron-induced damage—a key factor in determining whether a fusion reactor is economically viable.

We use *nucleonics* as a working name for this longer-term ambition. Just as electronics provides tools to engineer the behavior of electronic excitations, nucleonics would provide tools to engineer nuclear energy pathways. The fundamental challenge is that nuclear transitions typically involve energies in the keV–MeV range, whereas accessible electromagnetic, vibrational and magnetic controls operate at much lower energies and couple only weakly to most nuclear states. Recent advances in quantum engineering suggest possible ways to bridge this mismatch through resonances, collective states, interference and shared modes [1]. Theoretical and experimental progress in the field of nuclear quantum optics in particular suggest that we now have sufficient grounds to make a serious attempt at **establishing for nuclear systems a capability analogous to what electronics has achieved for electronic excitations.**

The proposed Challenge targets a foundational step toward this vision: demonstrating that nuclear energy can be transferred between distinct nuclear systems through an engineered, low-energy environment, with a quantitatively predictable and controllable outcome.

Fusion as a north-star application

The economic viability of fusion energy depends to a large extent on managing reaction products. Fast neutrons damage reactor components and force costly replacements and shutdowns. Whyte and co-workers [2] show that damage tolerance, replacement costs and downtime periods can become threshold constraints: in scenarios where neutron damage cannot be sufficiently mitigated, a fusion plant can be economically unviable even if it produces more energy than it consumes.

Lowering neutron damage could therefore be decisive in moving an otherwise nonviable plant into a commercially interesting regime. In an illustrative replacement-intensive D–D fusion scenario based on [2] (Appendix E), the effective electricity cost of the fusion plant is approximately 13.5 US cents/kWh, exceeding the assumed wholesale net electricity sale price of 10.0 US cents/kWh (following Whyte et al.'s representative base case). The economic gain factor is the ratio of net revenue to modeled electricity cost, with $Q_{\mathrm{econ}} = 1$ marking break-even. The plant would therefore operate at an economic loss ($Q_{\mathrm{econ}} \simeq 0.74$). Assuming unchanged useful power, reducing neutron emission by 50% leads to proportionally reduced damage and lowers the modeled electricity cost from 13.5 to 8.4 US cents/kWh, moving the plant above economic break-even ($Q_{\mathrm{econ}} \simeq 1.20$); a 90% extrapolative stretch scenario gives 4.2 US cents/kWh ($Q_{\mathrm{econ}} \simeq 2.37$). The economic rationale for suppressing neutron emission applies to a wide range of different fusion approaches.

In D–D fusion, a prospective nuclear energy routing implementation looks as follows: two deuterons enter a transient four-nucleon configuration that ordinarily decays through a neutron-producing channel about half the time. A sufficiently fast competing pathway could instead transfer its energy to a suitable nuclear acceptor system before that decay, enabling a more manageable acceptor-side outcome such as gamma emission.[1]

Fusion is a demanding long-term application that goes beyond the scope of a single four-year Challenge. A Challenge can, however, lay the cornerstone for this overall approach. The first step is to demonstrate and quantitatively predict controllable transfer of more than 5 keV of energy between distinct nuclear systems; later work addresses higher energies and shorter donor lifetimes.

[1] The relevant donor need not be the shortest-lived four-nucleon configuration: proposed 2+2 states of ^{4}He could provide longer-lived configurations in which such competition may be more favorable [3, 4].

Why this may now be possible – and what still needs to be solved

At molecular scales, energy can be routed from a donor into an acceptor with a more useful output. In CO_2 lasers, N_2 transfers excitation to CO_2, enabling infrared emission [5]. More relevant to nuclei, whose direct interactions are short-ranged, Cargioli *et al.* demonstrated externally switchable energy transfer between separated molecules through a shared optical-cavity mode [6]. The mode couples donors and acceptors without supplying the transferred excitation: energy is redirected into a system able to release it in a more useful form.

Moving the same idea to nuclei creates a much harder problem because nuclear transition energies are large and microscopic couplings to their surroundings are usually weak. But nuclei are not completely isolated from their environment, and several potentially helpful effects are already known. Groups of nuclei can behave collectively rather than independently, and coherent interaction between spatially separated sets of ^{57}Fe nuclei have been demonstrated; and nuclear dynamics have been manipulated using magnetic excitations, phase-controlled x-ray fields and mechanically engineered structures [7–12]. These experiments do not yet satisfy the low-energy nuclear-routing criterion targeted here, but they show that collective effects and engineered surroundings can be used to manipulate nuclear states.

A central trade-off is that using a low-energy mediator to transfer a much larger quantum of nuclear energy generally makes the transfer weak: the farther the mediator is from nuclear resonance, the smaller the effective coupling. Collective enhancement offers a possible counterweight. If many donor and acceptor nuclei participate coherently, their contributions to transfer can reinforce one another and potentially compensate for the suppression caused by large detuning. For an idealized single virtual pathway with phase-matched donor and acceptor ensembles, the effective transfer coupling scales as

$$J_{\mathrm{D-A}} \sim \frac{\sqrt{N_D N_A}\, g_D g_A}{\Delta}, \tag{1}$$

where N_D and N_A are the numbers of coherently participating nuclei, g_D and g_A are their individual couplings to the mediator, and Δ is the detuning. However, this simple expectation is not sufficient on its own: interference between transfer pathways, mode structure and other physical constraints can strongly limit or even eliminate the anticipated enhancement.

The challenge is to turn these competing scaling effects into a sufficiently strong and controllable transfer pathway. Appendix C explains obstacles in more detail and the main candidates to overcome them, including intermediate nuclear states, engineered loss, nonlinear coupling, electronic mediation and higher-frequency or hybrid environments.

Further motivation, albeit contingent on further substantiation, comes from experimental reports in which nuclear observables vary strongly with operating conditions in driven plasmas [13] and with beam energy or host material in deuteron-bombarded metal targets [3, 4, 14, 15]. Both strongly driven plasmas and condensed matter environments support collective electromagnetic, electronic and ionic modes and can naturally contain hydrogen species together with medium- and heavy-mass nuclear species, making them suitable environments for the above-mentioned mechanisms to play out under suitable conditions.

The Challenge design

The above considerations lead to the Challenge theme:

Can low-energy engineered controls route more than 5 keV of energy between nuclear systems—and can we predict and scale that transfer toward higher energies and shorter donor lifetimes?

The initial keV demonstration targets the most fundamental uncertainty: whether the necessary physical mechanisms can be combined into controllable nuclear energy-transfer pathways that work well enough to yield observable outcomes. Establishing this capability would establish an experimentally grounded development trajectory with testable scaling laws. Extending transfer toward MeV energies and shorter donor lifetimes remains formidable, but would then build on a strong foundation.

At the heart of the Challenge are three questions that teams must answer spanning mechanism, experiment and scaling:

1. What creates the transfer link between donor and acceptor nuclei? Teams must identify specific donor and acceptor states, the environment connecting them, and the microscopic interaction that enables energy transfer. Teams must also identify the physical effects that strengthen or control the transfer—for example resonance, collective behavior, nonlinear coupling, electronic mediation or engineered loss. They must build a quantitative model that predicts the absolute transfer rate from inputs and predicts how that rate responds to changes among the inputs.

2. Can the link be made strong enough for observable outcomes? Teams must reduce their model to a concrete experimental implementation, which can then be tested to validate or falsify the model. The resulting experiment must demonstrate transfer of more than 5 keV with acceptor-side $N_{\mathrm{day,det}} \geq 1$ detected event per 24 hours, while ruling out direct acceptor excitation, ordinary radiative transport and experimental background. At least one control parameter must be varied and produce a corresponding quantitatively predicted response.

3. How does it scale toward consequential nuclear control? Teams must determine how the mechanism changes as transition energies rise and donor lifetimes shorten. Beyond demonstrating an absolute transfer rate, they must establish how the engineered transfer rate scales relative to competing natural decay or loss pathways. The resulting scaling map must identify the dominant bottlenecks and engineering levers and define a concrete next rung: a higher-energy and/or shorter-lived nuclear system in which the same mechanism remains quantitatively testable. Fusion-relevant transfer is a long-term goal of this ladder.

Program structure

Over four years, teams compete through three evidence-gated phases: establish the mechanism, design and predict the experiment, then demonstrate and scale. Architectures requiring unsupported collective factors, implausible fine tuning or experimentally implausible material parameters do not advance.

Phase	Mission	Work and advancement gate
1: 0–12 mo Up to 8 teams EUR 0.75M/team	**Establish the mechanism**	Develop and compare competing physical routes. Deliver a quantitative model identifying a specific donor–acceptor coupling and a credible, detectable > 5 keV architecture. *Gate: advance up to 5 teams with viable mechanisms and experimentally credible predictions.*
2: 12–24 mo Up to 5 teams EUR 1.5M/team	**Design and predict the experiment**	Specify donor, acceptor, engineered environment, and readout. Constrain key model inputs independently; predict the absolute transfer rate and acceptor-side detected signal. *Gate: advance up to 3 teams whose experimental designs credibly predict* $N_{\mathrm{day,det}} \geq 1$.
3: 24–48 mo Up to 3 teams EUR 5M/team	**Demonstrate and scale**	Build and run the integrated experiment. Pass the common final exam by demonstrating attributable, controllable nuclear energy transfer consistent with the pre-experiment prediction. Validate a scaling relation and define a higher-energy and/or shorter-lifetime next test.

The final exam: Start with a nuclear donor system carrying at least 5 keV of nuclear energy. Enable the engineered transfer pathway and show excitation appearing in a distinguishable acceptor state or manifold with $N_{\mathrm{day,det}} \geq 1$ detected acceptor-side event per 24 hours attributable to mediated transfer. Switch, detune or otherwise modulate the engineered environment or its coupling to the nuclei, and show that the acceptor-side signal changes by the predicted amount. Infer the transfer rate using calibrated efficiencies, compare it with the pre-experiment prediction, and demonstrate at least one controlled scaling relation toward a named higher-energy and/or shorter-lifetime next step.

Maximum program envelope: EUR 28.5M over four years. The staged funnel concentrates resources progressively: A total of EUR 6M to establish the most credible physical links, a further EUR 7.5M to turn the strongest concepts into quantitative experiments, and EUR 15M for up to three integrated demonstrations.

Why Germany, why SPRIND

Germany already has much of the required scientific stack. The Max Planck Institute for Nuclear Physics works on nuclear quantum dynamics and coherent control of nuclei [16]; GSI/FAIR on nuclear structure, reactions and spectroscopy [17]; DESY's PETRA III P01 provides dedicated nuclear-resonant-scattering infrastructure [18]; and the Pálffy-Buß group in Würzburg works on nuclear quantum optics, nuclear excitation and collective dynamics [19]. Germany's wider quantum, materials, plasma and strong-field communities add expertise in cavities, coherent control, phonons, magnons, plasma modes and many-body dynamics.

SPRIND can convene capabilities spread across institutional mandates and apply a forcing function: compare mechanisms under common gates, stop approaches without credible transfer-rate predictions, and carry successful physics toward repeatable technology, intellectual property and company formation. This opportunity is too interdisciplinary and uncertain for conventional research funding, yet consequential enough for a concentrated, milestone-driven SPRIND Challenge. Such a Challenge would allow for testing the physics before committing substantial resources downstream.

Success would establish a foundational nuclear-energy-routing primitive for nucleonics, alongside trained teams, a mapped design space and a shared technology agenda. The ambition is for **Germany, and Europe, to pioneer a new scientific field and the industries built around it**, with multiple leading companies.

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Appendix A: Measurement and validation criteria

A successful experiment must rule out direct acceptor excitation, ordinary radiative transfer from donor to acceptor and other backgrounds. The experimental geometry and controls must allow these channels to be suppressed or independently quantified. Spatial separation and shielding are preferred where compatible with the proposed mechanism, but are not themselves requirements.

The experiment must convert the observed acceptor-side signal—for example detected particles or photons—into an estimate of the number of transferred nuclear excitations, correcting for detector efficiency, geometric acceptance, branching ratios, live time and other independently calibrated losses. Together with the measured donor population and exposure time, this yields an experimentally inferred transfer rate.

Minimum observable signal. The decisive experiment must achieve $N_{\mathrm{day,det}} \geq 1$, where $N_{\mathrm{day,det}}$ is the number of detected acceptor-side events per 24 hours attributable to excitation of the acceptor through the engineered transfer pathway. The signal must also be statistically distinguishable from backgrounds and relevant control measurements.

Teams must predict $N_{\mathrm{day,det}}$ before the integrated experiment from independently measured or constrained system parameters, and the observed rate must be consistent with that prediction within stated uncertainties. Teams must also vary at least one pre-specified control variable—for example drive amplitude, detuning, loss, environment structure or the number of coherently participating nuclei—and recover the predicted change in transfer.

The objectives are to detect nuclear excitation transfer, to show that it is quantitatively understood, and that it is controllable.

Appendix B: Draft Challenge specification

Nuclear donor and acceptor. The initial energy must reside in a nuclear excitation, and the receiver signal must identify excitation of a distinct nuclear acceptor system. The receiver may be a single nuclear transition or a distinguishable multi-nucleus or collective final state.

Qualifying low-energy transfer. A qualifying demonstration must transfer > 5 keV of nuclear excitation from an initially excited nuclear donor to a distinct nuclear acceptor, while all externally applied control or mediator quanta that facilitate the transfer have energies at least 10^3 smaller than the transferred nuclear energy.

Radiation or other fields used solely to prepare the donor or read out the acceptor do not count toward this limit, provided they do not facilitate the donor-to-acceptor transfer itself.

Engineered transfer pathway. The transfer must be genuinely enabled or controlled by an engineered electromagnetic, phononic, magnonic, electronic, plasma or hybrid environment. The environment must enter the donor-to-acceptor transfer dynamics; it cannot merely heat the system or directly supply the transferred keV-scale nuclear energy. The proposed mechanism must identify the physical donor-to-acceptor coupling and specify an experimental change that should suppress, detune or otherwise alter the transfer.

Control requirement. The engineered contribution must be experimentally switchable, detunable or otherwise modulated so that it can be distinguished from fixed backgrounds, direct acceptor excitation and ordinary donor decay.

Appendix C: Known constraints on linear shared-mode transfer and prospective escape routes

A generic reference model serves as a starting point to lay out known constraints and escape routes.

For donor and acceptor transitions coupled linearly to a harmonic mediator,

$$H = \sum_j \frac{E_j}{2}\sigma_j^z + \hbar\omega b^\dagger b + \sum_j g_j X_j(\eta b + \eta^* b^\dagger).$$

Constraint 1 — a slow linear bus can nearly cancel its own exchange. For resonant donor and acceptor transition energy E,

$$J_{\mathrm{lin}} = g_D g_A \left[\frac{1}{E - \hbar\omega} - \frac{1}{E + \hbar\omega} \right] \simeq \frac{2\hbar\omega}{E} \frac{g_D g_A}{E}.$$

The two virtual path families enter with opposite signs. A rotating-wave estimate that keeps only one denominator can therefore strongly overestimate exchange when $\hbar\omega \ll E$ [20, 21]. Path-dependent loss or doorway structure can break this cancellation, but that effect must emerge from the relevant open-channel dynamics rather than be inserted as an unconstrained enhancement.

Constraint 2 — driving a linear mediator harder does not by itself strengthen the inter-nuclear link. With a coherent displacement $b = \alpha(t) + c$, the interaction separates into a classical local drive and a quantum fluctuation term,

$$H_{\mathrm{int}} = \sum_j f_j(t) X_j + \sum_j g_j X_j (\eta c + \eta^* c^\dagger).$$

The first term can dress or drive each nucleus locally; the second carries the connected quantum exchange between different nuclei and is independent of the coherent amplitude α. Coherent drive can therefore enhance a genuine Raman or nonlinear vertex, but it cannot simply replace an inter-site zero-point coupling by $g\sqrt{n}$ in a linear harmonic bus.

Constraint 3 — collective enhancement is limited by mode normalization. Collective states can in principle enhance energy transfer; for example, Lloyd and Mohseni predict "supertransfer" between symmetrized donor and acceptor states [22]. Whether such enhancement survives in a specific nuclear architecture, however, depends on how the collective states couple to the mediating mode. For a single normalized extended mode, increasing the number of participating nuclei increases the collective bright-state factor but decreases the zero-point motion associated with each site. As a result, large isotope inventory alone does not imply an arbitrarily large donor-to-acceptor coupling. Any claimed collective gain must be calculated after mode normalization, spectral participation and phase matching are included.

Four known escape routes:

- **Pathway asymmetry, engineered loss and doorway states.** Make competing virtual pathways physically inequivalent so that the leading cancellation no longer applies.
- **Nonlinear, Raman or Floquet links.** Use a microscopic interaction in which an applied drive genuinely modifies the donor-to-acceptor coupling rather than merely increasing occupation of an otherwise linear mediator.
- **Electronic and alternative-mode links.** Electronic bridges, plasmons, optical or local modes, cavities and plasma modes can provide different selection rules and larger microscopic fields while still satisfying the Challenge's low-energy-routing criterion [23].
- **Alternative collective or non-bosonic links.** Collective nuclear eigenstates, delocalized mechanical structures, neutron transfer or other mechanisms may evade assumptions of the single linear-boson model. Proposals must state explicitly which assumption they break.

Two common metrics help distinguish an interesting enhancement from a useful absolute coupling:

$$\mathcal{E} = \frac{J_{\mathrm{predicted}}}{J_{\mathrm{lin}}}, \qquad \mathcal{V} = \frac{J_{\mathrm{predicted}}}{J_{\mathrm{required}}}.$$

$\mathcal{E}$ asks whether an architecture beats the linear-bus reference; $\mathcal{V}$ asks whether it does so by enough to compete with the relevant nuclear lifetime or branching process. Phase 1 proposals should report both wherever possible.

Appendix D: Empirical boundary cases and hypothesis-generating observations

The Challenge does not depend on anomalous nuclear claims being correct. A small and heterogeneous literature nevertheless provides useful boundary cases for asking whether nuclear outputs can depend on material state or external drive more strongly than simple models capture. These observations should be treated according to evidentiary strength and converted into controlled parameter-mapping experiments rather than used as premises of the program.

Higher-confidence boundary case: enhanced D–D fusion rates in beam-target fusion configurations. Karahadian *et al.* report a pronounced low-energy fusion-yield plateau in Pd and Ti hydrides below about 2 keV, with conventional D–D products directly detected [14]. This does not establish excitation transfer, but it provides a modern platform in which nuclear output can be mapped against material state and beam conditions.

Host-dependent conventional nuclear observables. Huke *et al.* reported altered neutron anisotropy and n/p branching for Li, Sr and Na below about 20 keV, while emphasizing the experimental complications of these reactive targets [15]. Because the n/p ratio

measures which exit channel the reaction takes, host dependence of this ratio is directly relevant to whether the environment can influence neutron-producing decay.

Alternative D–D reaction products. Dubey *et al.* [4] report experimental signatures of an internal electron–positron pair-creation channel in low-energy D–D fusion, interpreted through a proposed near-threshold 0^+ resonance. This provides motivation for investigating nuclear exit channels beyond the familiar neutron- and proton-producing branches.

Exploratory drive-correlated claims. Fomitchev-Zamilov reports neutron emission coincident with acoustic cavitation of deuterated titanium powder while noting that alternative processes still require exclusion [24]. Klimov and collaborators report neutron and soft-x-ray signals from pulsed heterogeneous plasmoids [13]. These nuclear claims require independent validation. Programmatically, their value is to motivate experiments that measure the relevant acoustic or plasma state simultaneously with nuclear outputs and ask whether any signal follows a specific collective mode or operating regime rather than deposited power alone.

Appendix E: Illustrative economic sensitivity to neutron reduction

We adapt the concept-agnostic framework of Whyte et al. [2] to a hypothetical, replacement-intensive D–D plant. Their Appendix B considers D–D fusion without subsequent tritium or ^{3}He burning: 2.45 MeV neutrons carry $f_{n,0} \simeq 0.34$ of fusion energy. With a neutron-damage factor $F_{\mathrm{dpa}} \simeq 16$, an illustrative 25-displacements-per-atom (dpa) damage limit would yield a surface energy-fluence limit of $25/(0.34 \times 16) \simeq 4.6$ MW y/m^2. Here we take $X_{S,0} = 3.125$ MW y/m^2, Whyte's Table 2 central value (motivated by D–T). Applied to this D–D scenario, that corresponds to an illustrative effective damage limit of about 17 dpa. X_S represents cumulative *fusion-energy throughput* per unit surface before damage requires replacement. For a fractional reduction r in neutron-induced damage *per unit fusion energy*, we assume an unchanged residual neutron spectrum and neutron-limited component lifetime, giving $X_S(r) = X_{S,0}/(1-r)$. Equivalent reductions in neutron emission have this effect if useful fusion output and damage per remaining neutron are preserved.

We assume areal fusion power $w = P_f/S = 4$ MW/m^2, net conversion efficiency $\eta_E = 0.40$, replacement downtime $\tau_{\mathrm{rep}} = 0.30$ y, replacement cost $c_{\mathrm{rep}} = 1$ M\$/m^2, plant cost $c_{\mathrm{FPP}} = 10$ M\$/m^2, plant life $T = 30$ y and real financing rate $i = 0.02$. The replacement assumptions lie within the ranges explored by Ref. [2]. Following its representative base case, we assume net electricity revenue $p_{\mathrm{net}} = 100$ \$/MWh (10.0 US cents/kWh), motivated by approximately 160 \$/MWh US retail pricing, generation accounting for roughly two-thirds, and variable operating costs already accounted for; target/consumable cost is set to zero. This is an illustrative net generator-revenue benchmark, not an observed wholesale price or universal break-even threshold.

With annual costs C in M\$/m^2 y, Eqs. (2), (6), (17), (20), (29) and (40)–(41) of Ref. [2] give

$$\begin{gathered}
\tau_{\mathrm{op}} = X_S/w, \qquad U = \frac{\tau_{\mathrm{op}}}{\tau_{\mathrm{op}} + \tau_{\mathrm{rep}}}, \qquad C_{\mathrm{rep}} = \frac{c_{\mathrm{rep}}}{\tau_{\mathrm{op}} + \tau_{\mathrm{rep}}}, \\
C_{\mathrm{fixed}} = c_{\mathrm{FPP}} \frac{i(1+i)^T}{(1+i)^T - 1} = 0.4465, \qquad L_{\mathrm{eff}} = \frac{10^6 (C_{\mathrm{fixed}} + C_{\mathrm{rep}})}{8760 w \eta_E U} \\
= 31.86 + 103.56(1-r) \text{ \$/MWh} = 3.186 + 10.356(1-r) \text{ US cents/kWh}, \\
Q_{\mathrm{econ}} = p_{\mathrm{net}}/L_{\mathrm{eff}}.
\end{gathered} \tag{2}$$

The simple linear dependence on r follows from $1/U = 1 + w\tau_{\mathrm{rep}}/X_S$ and $C_{\mathrm{rep}}/U = c_{\mathrm{rep}} w/X_S$: both damage-related contributions to L_{eff} scale as $1/X_S \propto (1-r)$. This is a consequence of the assumed replacement-limited model. Physically, less damage means fewer replacements and higher utilization; the latter spreads fixed costs over more electricity sold. The resulting sensitivity is:

	No reduction	50% (base)	90% (stretch)
X_S (MW y/m^2)	3.125	6.25	31.25
τ_{op} (y)	0.78	1.56	7.81
Utilization U	72.3%	83.9%	96.3%
C_{rep} (M\$/m^2 y)	0.925	0.537	0.123
L_{eff} (US cents/kWh)	13.54	8.36	4.22
Q_{econ}	0.74	1.20	2.37

At the assumed net revenue of 10.0 US cents/kWh, $Q_{\mathrm{econ}} = 1$ marks modeled break-even: the baseline operates at a loss, whereas 50% damage reduction lowers electricity cost from 13.5 to 8.4 US cents/kWh, crossing break-even at $r \simeq 34.2\%$. The operating interval doubles, annual replacement expenditure falls from 0.925 to 0.537 M\$/m^2 y, and utilization rises from 72.3% to 83.9%. The 50% case remains below the paper's illustrative $Q_{\mathrm{econ}} = 1.5$ commercial screening benchmark, crossed at $r \simeq 66.4\%$; the 90% stretch case gives 4.2 US cents/kWh ($Q_{\mathrm{econ}} \simeq 2.37$). The screening benchmark is not an investment guarantee. Under baseline assumptions, increasing fusion power density w alone cannot reach even break-even; Eq. (44) yields $Q_{\mathrm{econ,max}} \simeq 0.97$ at $r = 0$.

These scenarios hold useful power and other costs constant. Both reduced-damage cases extend beyond the paper's surveyed X_S range (up to 6 MW y/m^2). In particular, the 90% case implies only about 3.7 replacement cycles over 30 years, weakening the steady-cycle approximation. Specific designs require detailed neutronics and systems costing.

References

1. F. Metzler, M. K. Lilley, M. C. DeCapua, P. L. Hagelstein, J. F. Messinger, and N. Galvanetto, "The next phase of quantum energy science: from mechanisms to applications," (2026). https://arxiv.org/abs/2610.27165., arXiv preprint arXiv:2610.27165 [quant-ph].

2. D. G. Whyte, A. Lo, R. Bielajew, M. Hancock, R. Moeykens, and G. Shaw, "Criteria for the economic viability of fusion power plants," *Journal of Fusion Energy*, **45**, 49 (2026). doi:10.1007/s10894-026-00577-9.

3. K. Czerski, "Deuteron-deuteron nuclear reactions at extremely low energies," *Physical Review C*, **106**, *1*, L011601 (2022). doi:10.1103/PhysRevC.106.L011601.

4. R. Dubey, K. Czerski, G. Das H., A. Kowalska, N. Targosz-Sleczka, M. Kaczmarski, and M. Valat, "Experimental Signatures of a New Channel of the Deuteron-Deuteron Reaction at Very Low Energy," *Physical Review X*, **15**, *4*, 041004 (2025). doi:10.1103/chlp-b215.

5. C. K. N. Patel, "Selective Excitation through Vibrational Energy Transfer and Optical Maser Action in N2–CO2," *Physical Review Letters*, **13**, *21*, 617 (1964). doi:10.1103/PhysRevLett.13.617.

6. A. Cargioli, M. Lednev, L. Lavista, A. Camposeo, A. Sassella, D. Pisignano, A. Tredicucci, F. J. Garcia-Vidal, J. Feist, and L. Persano, "Active control of polariton-enabled long-range energy transfer," *Nanophotonics*, **13**, *14*, 2541 (2024). doi:10.1515/nanoph-2023-0677.

7. R. Röhlsberger, K. Schlage, B. Sahoo, S. Couet, and R. Rüffer, "Collective Lamb shift in single-photon superradiance," *Science*, **328**, 1248 (2010). doi:10.1126/science.1187770.

8. A. I. Chumakov, A. Q. R. Baron, I. Sergueev, C. Strohm, O. Leupold, Y. Shvyd'ko, G. V. Smirnov, R. Rüffer, Y. Inubushi, M. Yabashi, K. Tono, T. Kudo, and T. Ishikawa, "Superradiance of an ensemble of nuclei excited by a free electron laser," *Nature Physics*, **14**, 261 (2018). doi:10.1038/s41567-017-0001-z.

9. J. Haber, X. Kong, C. Strohm, S. Willing, J. Gollwitzer, L. Bocklage, R. Rüffer, A. Pálffy, and R. Röhlsberger, "Rabi oscillations of X-ray radiation between two nuclear ensembles," *Nature Photonics*, **11**, 720 (2017). doi:10.1038/s41566-017-0013-3.

10. L. Bocklage, J. Gollwitzer, C. Strohm, C. F. Adolff, K. Schlage, I. Sergeev, O. Leupold, H.-C. Wille, G. Meier, and R. Röhlsberger, "Coherent control of collective nuclear quantum states via transient magnons," *Science Advances*, **7**, eabc3991 (2021). doi:10.1126/sciadv.abc3991.

11. K. P. Heeg, A. Kaldun, C. Strohm, C. Ott, R. Subramanian, D. Lentrodt, J. Haber, H.-C. Wille, S. Goerttler, R. Rüffer, C. H. Keitel, R. Röhlsberger, T. Pfeifer, and J. Evers, "Coherent X-ray–optical control of nuclear excitons," *Nature*, **590**, 401 (2021). doi:10.1038/s41586-021-03276-x.

12. S. Velten, L. Bocklage, X. Zhang, K. Schlage, A. Panchwanee, S. Sadashivaiah, I. Sergeev, O. Leupold, A. I. Chumakov, O. Kocharovskaya, and R. Röhlsberger, "Nuclear quantum memory for hard x-ray photon wave packets," *Science Advances*, **10**, eadn9825 (2024). doi:10.1126/sciadv.adn9825.

13. A. I. Klimov, N. K. Belov, and B. N. Tolkunov, "Neutron Flux and Soft X-Radiation Created by Heterogeneous Plasmoid," *Journal of Physics: Conference Series*, **1698**, 012034 (2020). doi:10.1088/1742-6596/1698/1/012034.

14. M. E. Karahadian, M. Colborne, A. Persaud, T. Schenkel, and J. N. Munday, "Enhanced nuclear fusion in the sub-keV energy regime," *Nature Communications* (2026). doi:10.1038/s41467-026-74421-1.

15. A. Huke, K. Czerski, T. Dorsch, A. Biller, P. Heide, and G. Ruprecht, "Evidence for a host-material dependence of the n/p branching ratio of low-energy d+d reactions within metallic environments," *European Physical Journal A*, **27**, *S1*, 187 (2006). doi:10.1140/epja/i2006-08-028-3.

16. Max Planck Institute for Nuclear Physics, "X-ray nuclear quantum dynamics," Research overview, Correlated and X-Ray Quantum Dynamics group (2026). https://www3.mpi-hd.mpg.de/mpi/en/forschung/abteilungen-und-gruppen/theoretische-quantendynamik-und-quantenelektrodynamik/forschung., accessed September 2026.

17. GSI Helmholtzzentrum fuer Schwerionenforschung and FAIR, "Research fields and experiments: nuclear structure, astrophysics and reactions," Institutional research overview (2026). https://www.gsi.de/en/work/research., accessed September 2026.

18. Deutsches Elektronen-Synchrotron DESY, "P01 High Resolution Dynamics Beamline and Nuclear Resonant Scattering Station," PETRA III Photon Science facility description (2026). https://photon-science.desy.de/e58/e176720/e177229/e177261/e177290/index_eng.html., accessed September 2026.

19. Palffy-Buss Group, Julius-Maximilians-Universitaet Wuerzburg, "Research topics: X-ray quantum optics and nuclear quantum dynamics," Research group overview (2026). https://www.physik.uni-wuerzburg.de/research-groups/palffy-buss-group/research-topics/., accessed September 2026.

20. P. L. Hagelstein and I. U. Chaudhary, "Excitation transfer in two two-level systems coupled to an oscillator," *Journal of Physics B: Atomic, Molecular and Optical Physics*, **41**, 135501 (2008). doi:10.1088/0953-4075/41/13/135501.

21. R. G. DeLosh and W. J. C. Grant, "Electronic Energy Transfer via Virtual-Phonon Processes," *Physical Review B*, **1**, *4*, 1754 (1970).

22. S. Lloyd and M. Mohseni, "Symmetry-enhanced supertransfer of delocalized quantum states," *New Journal of Physics*, **12**, 075020 (2010). doi:10.1088/1367-2630/12/7/075020.

23. S. G. Porsev, V. V. Flambaum, E. Peik, and C. Tamm, "Excitation of the isomeric ^{229m}Th nuclear state via an electronic bridge process in ^{229}Th^{+}," *Physical Review Letters*, **105**, 182501 (2010). doi:10.1103/PhysRevLett.105.182501.

24. M. Fomitchev-Zamilov, "Observation of neutron emission during acoustic cavitation of deuterated titanium powder," *Scientific Reports*, **14**, 11517 (2024). doi:10.1038/s41598-024-62055-6.